

GenieMart

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Project Definition

University students are always shopping for furniture, clothes, textbooks and other materials.

Platforms like Facebook Marketplace and Craigslist that aren't directly geared towards college communities are prone to more scams, unreliable sellers and safety hazards. GenieMart will be geared only towards UNCG students/staff and helping the UNCG community. A platform only accessible to the UNCG community will help ensure safer transactions, more reliable sellers being that the sellers are known to be part of the UNCG community and will help give back to the people of the university. With GenieMart, any UNCG student or person who owns an active UNCG email will be able to put items up for sale that other UNCG members can look at, purchase, comment and favorite. GenieMart will hopefully be a helpful and useful tool for UNCG students and staff to be able to give back to one another in a safer and more reliable way. GenieMart will be a mobile app which helps with convenience and quickness. The architecture of GenieMart will consist of React Native frontend (Expo), Python backend, and a PostgreSQL database.

Project Requirements Analysis

Functionality:

Users will be able to create an account with their uncg email, create listings for items, users can engage sellers through an in-app message board, view and like listings, and leave comments on sellers and items posted.

UI Usability:

Cross-platform mobile application, e-commerce type for item listings, and instant messaging for users and listers to engage, and a user dashboard (role-based features)

Security:

GenieMart is strictly for UNCG students; security is ensured by only allowing @uncg.edu domain emails to create an account, on top of verifying student status by having users upload pictures of their student identification card before buying and listing features are granted. Student identification cards are stored temporarily, and then the system will delete them after the verification process occurs.

Project Specification

Focus/Domain/Area:

The focus of our project is to provide the UNCG community with an online marketplace that enables students to safely buy and sell items within their campus community using university email authentication.

Libraries/Frameworks/Development:

Frontend Framework:

- React Native
 - Expo (framework)
 - Expo-router: navigation
 - Tailwind CSS: Styling/UI comments

Backend Framework:

- Python
 - Web framework:
 - FastAPI: Routes, HTTP requests, and API endpoints
 - Uvicorn: run FastAPI
 - Data Persistence:
 - PostgreSQL
 - ORM: SQLAlchemy
 - Authentication:
 - JWT(JSON Web Tokens)
 - Python-jose: to handle auth for each request
 - Passlib: password hashing
 - JWT will handle the role and role-based services
 - ID upload and Listing Pictures Upload:
 - python-multipart

Platform:

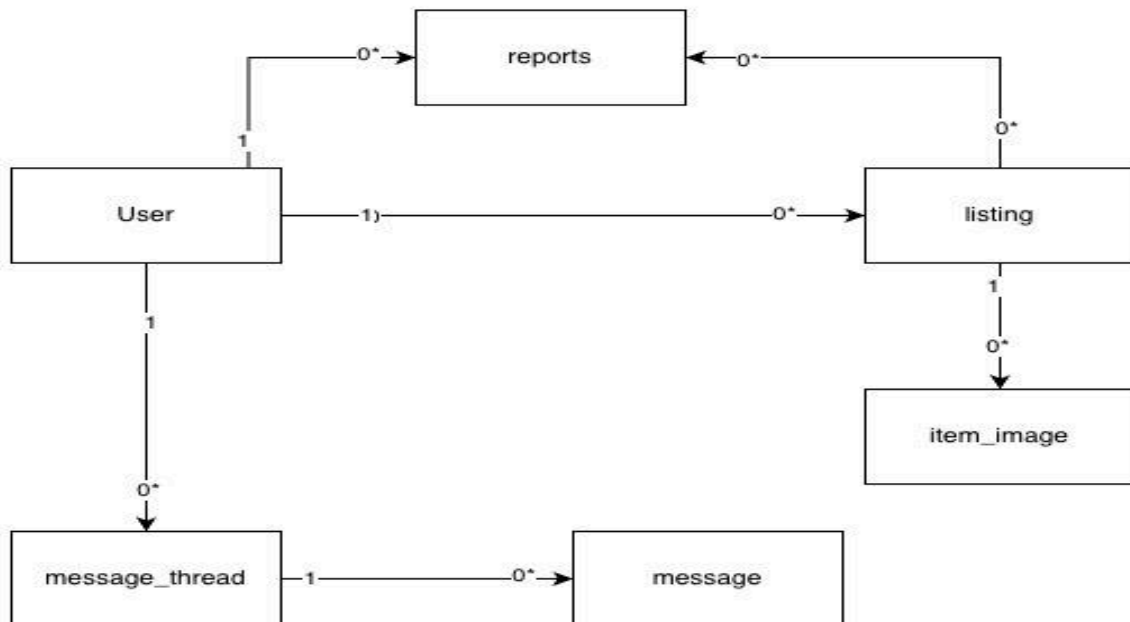
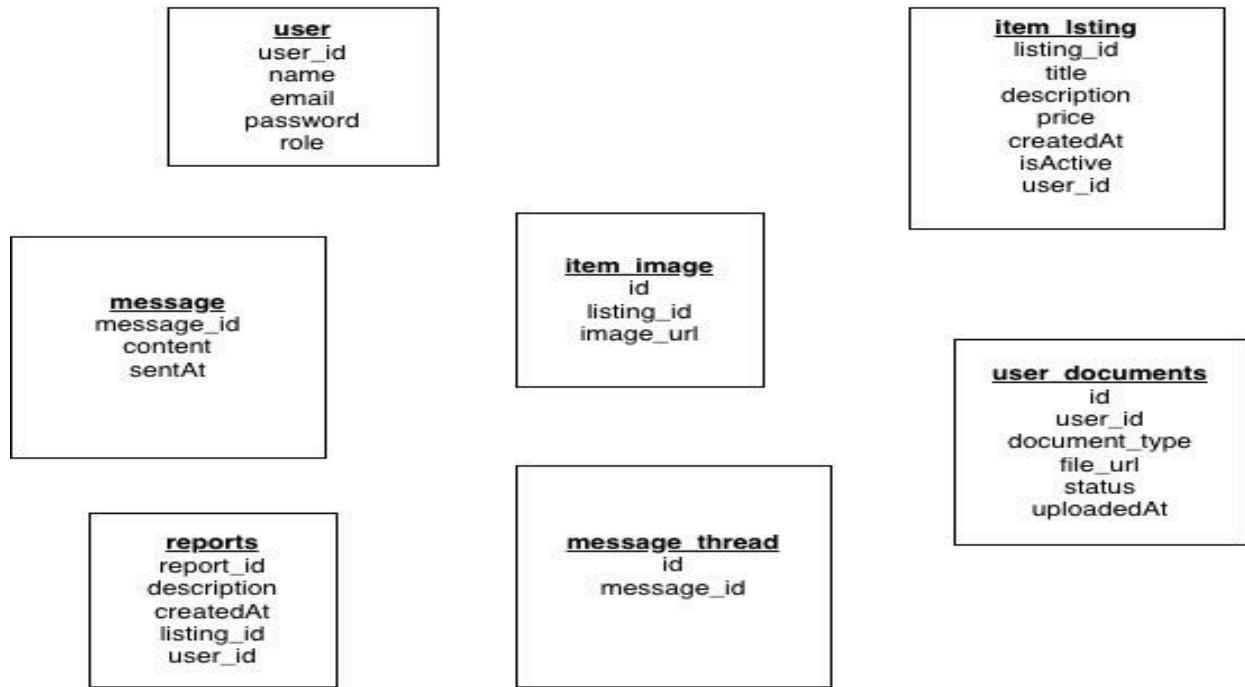
- Mobile

Genre:

- Online Multi-Vender Marketplace

System Design

PROPOSED EMTITIES



GenieMart
An Online Storefront

